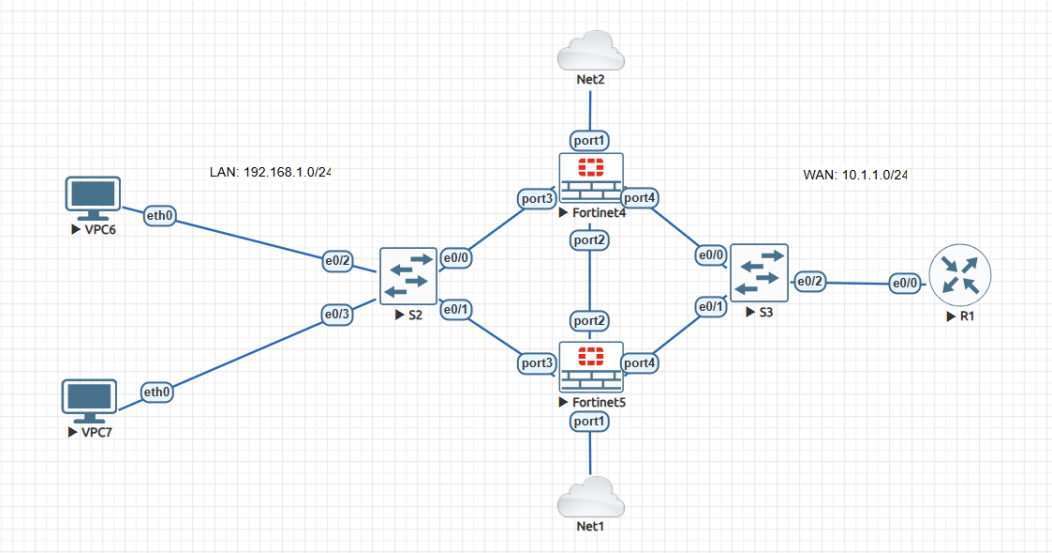
Objectifs

Déployer une architecture en Haute Disponibilité (HA) est un choix stratégique fondamental pour toute entreprise. Le déploiement en HA consiste à doubler les équipements critiques pour qu'en cas de panne du matériel principal, un équipement secondaire prenne le relais automatiquement et instantanément sans interruption des trafics**.**

Le choix d'un tel déploiement peut viser :

- Garantir la continuité d'activité
- Absorber les pannes lors des mises à jour
- Load Balancing
- Répondre aux exigences de conformité
- …

Topology

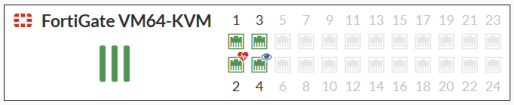


Router config

```
!
interface Loopback0
 ip address 8.8.8.8 255.255.255.0
!
interface Ethernet0/0
 ip address 10.1.1.1 255.255.255.0
!
```

Fortigate Configs

Physical Interface 4				
LAN (port3)	Physical Interface		192.168.1.2/255.255.255.0	PING HTTPS FMG-Access
port1	Physical Interface		192.168.91.140/255.255.255.0	PING HTTPS SSH HTTP FMG-Access
port2	Physical Interface		0.0.0.0/0.0.0.0	
WAN (port4)	Physical Interface		10.1.1.2/255.255.255.0	PING



FW1 (Primary)

Refresh | Edit | Remove device from HA cluster

Status	Priority	Hostname	Serial No.	Role	System Uptime	Sessions	Throughput
Synchronized	138	FW1	FGVMEVERNXB2PX44	Primary	55m 49s	18	62.00 kbps
Synchronized	128	FW2	FGVMEVHT-7P2DL77	Secondary	55m 51s	1	22.00 kbps

```
FW1 # show system int
name    Name.
fortilink   static   0.0.0.0 0.0.0.0  10.255.1.1 255.255.255.0  up   disable   aggregate
l2t.root   static   0.0.0.0 0.0.0.0  0.0.0.0 0.0.0.0  up   disable   tunnel
naf.root   static   0.0.0.0 0.0.0.0  0.0.0.0 0.0.0.0  up   disable   tunnel
port1   dhcp   0.0.0.0 0.0.0.0  192.168.91.140 255.255.255.0  up   disable   physical
port2   static   0.0.0.0 0.0.0.0  0.0.0.0 0.0.0.0  up   disable   physical
port3   static   0.0.0.0 0.0.0.0  192.168.1.2 255.255.255.0  up   disable   physical
port4   static   0.0.0.0 0.0.0.0  10.1.1.2 255.255.255.0  up   disable   physical
ssl.root   static   0.0.0.0 0.0.0.0  0.0.0.0 0.0.0.0  up   disable   tunnel
```

```
FW2 # show syst interface
name    Name.
fortilink   static   0.0.0.0 0.0.0.0  10.255.1.1 255.255.255.0  up   disable   aggregate
l2t.root   static   0.0.0.0 0.0.0.0  0.0.0.0 0.0.0.0  up   disable   tunnel
naf.root   static   0.0.0.0 0.0.0.0  0.0.0.0 0.0.0.0  up   disable   tunnel
port1   dhcp   0.0.0.0 0.0.0.0  192.168.91.140 255.255.255.0  up   disable   physical
port2   static   0.0.0.0 0.0.0.0  0.0.0.0 0.0.0.0  up   disable   physical
port3   static   0.0.0.0 0.0.0.0  192.168.1.2 255.255.255.0  up   disable   physical
port4   static   0.0.0.0 0.0.0.0  10.1.1.2 255.255.255.0  up   disable   physical
ssl.root   static   0.0.0.0 0.0.0.0  0.0.0.0 0.0.0.0  up   disable   tunnel
```

Tests

```
VPCS> show ip

NAME        : VPCS[1]
IP/MASK     : 192.168.1.6/24
GATEWAY     : 192.168.1.2
DNS         : 192.168.91.2
DHCP SERVER : 192.168.1.2
DHCP LEASE  : 603424, 604800/302400/529200
MAC         : 00:50:79:66:68:07
LPORT       : 20000
RHOST:PORT  : 127.0.0.1:30000
MTU         : 1500

VPCS> ping 8.8.8.8

84 bytes from 8.8.8.8 icmp_seq=1 ttl=254 time=1.977 ms
84 bytes from 8.8.8.8 icmp_seq=2 ttl=254 time=2.050 ms
84 bytes from 8.8.8.8 icmp_seq=3 ttl=254 time=1.587 ms
84 bytes from 8.8.8.8 icmp_seq=4 ttl=254 time=2.161 ms
84 bytes from 8.8.8.8 icmp_seq=5 ttl=254 time=2.030 ms
```

```
NAME        : VPCS[1]
IP/MASK     : 192.168.1.5/24
GATEWAY     : 192.168.1.2
DNS         : 192.168.91.2
DHCP SERVER : 192.168.1.2
DHCP LEASE  : 603425, 604800/302400/529200
MAC         : 00:50:79:66:68:06
LPORT       : 20000
RHOST:PORT  : 127.0.0.1:30000
MTU         : 1500

VPCS> ping 8.8.8.8

84 bytes from 8.8.8.8 icmp_seq=1 ttl=254 time=2.931 ms
84 bytes from 8.8.8.8 icmp_seq=2 ttl=254 time=1.518 ms
84 bytes from 8.8.8.8 icmp_seq=3 ttl=254 time=1.768 ms
84 bytes from 8.8.8.8 icmp_seq=4 ttl=254 time=1.218 ms
84 bytes from 8.8.8.8 icmp_seq=5 ttl=254 time=1.923 ms
```

FW2 (Primary)

Refresh | Edit | Remove device from HA cluster

Status	Priority	Hostname	Serial No.	Role	System Uptime	Sessions	Throughput
Synchronized	148	FW2	FGVMEVHT-7P2DL77	Primary	1h 3m	49	74.00 kbps
Synchronized	138	FW1	FGVMEVERNXB2PX44	Secondary	1h 3m	45	18.00 kbps